AGI - Benchmarks, Feasibility, Implications

Frederik Schittny, Institute of Theoretical Informatics, KIT
16.02.2026

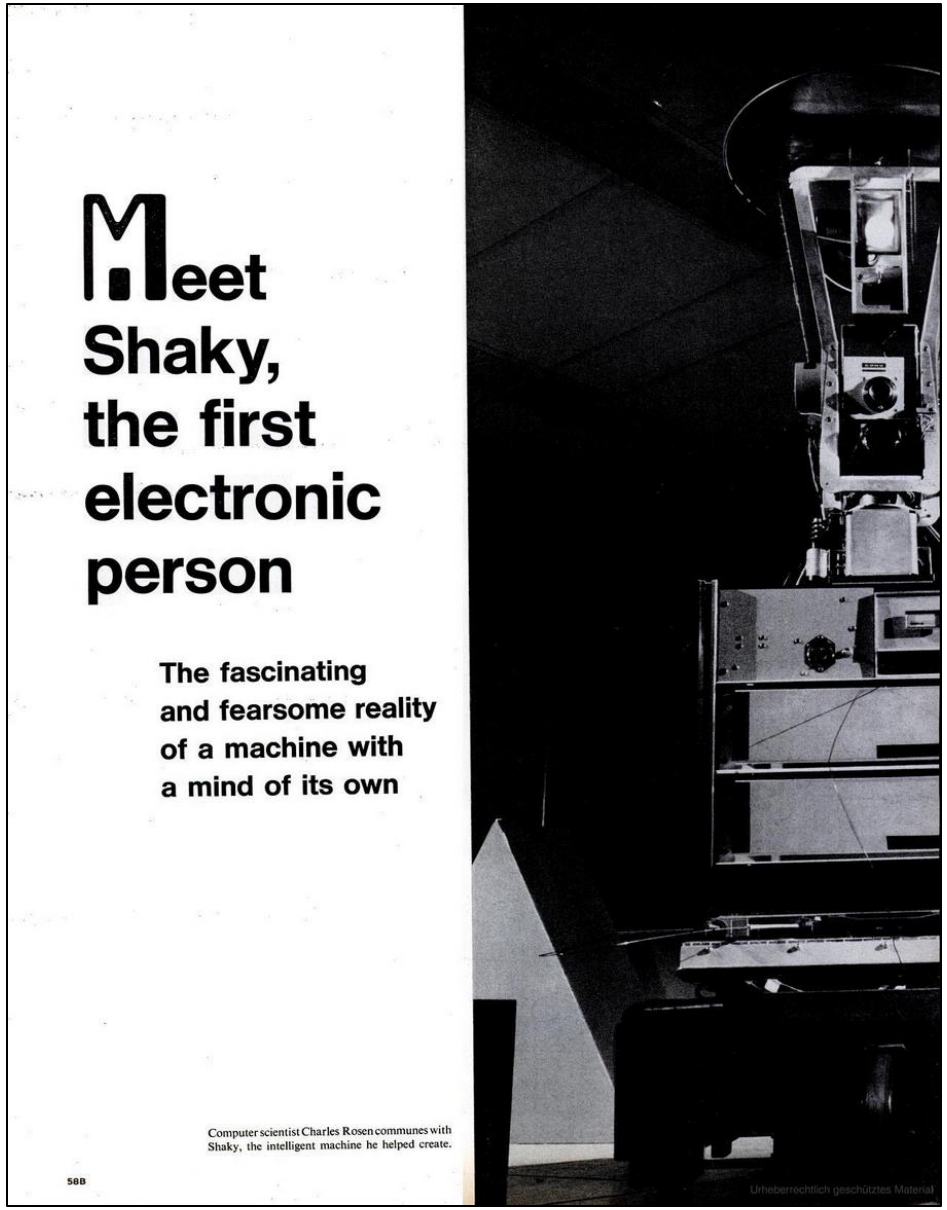

Sam Altman
@sama
In from three to eight years we will have a machine with the general intelligence of an average human being
8:11 PM · Mar 31, 2025 · 5.5M Views
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Marvin Minsky
@marv_minsky
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– Marvin Minsky, *Life* Magazine, November 1970



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AGI

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Artificial

AGI

Artificial General


AGI
Intelligence ???
Artificial
General

Defining Intelligence (is really hard)

Defining Intelligence (is really hard)

internal processes

behavior

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internal processes

behavior

rationality

“human”

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internal processes

behavior

psychology

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self-awareness

Defining Intelligence (is really hard)

emotions

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self-awareness

The most important question to keep in mind when talking about AGI:

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What is the underlying definition of AGI in this context?

The Turing Test

- Indistinguishability in conversation
  - Beat to various degrees
  - Language models “specialized” for this test
- Critcism
  - Language capabilities only part intelligence
  - Indistinguishability no prove of intelligence

Underlying intelligence understanding:

Indistinguishability between human and AI (in language use)



Humanities Last Exam

- 2500 expert level questions
  - Mostly scientific
  - “Multimodal” (text and images)
- Intended for foundation models
- Traditional LLM benchmarks “too easy”
  - Solution: make the questions harder

Underlying intelligence understanding:

Expert level knowledge and reasoning skills across many topics

ARC-AGI

- Measuring fluid intelligence
- Static puzzles
  - Somewhat comparable to IQ tests
  - Symbolic, compositional and contextual reasoning
- Interactive puzzles
  - Small games (without given rules)
  - Goal acquisition, planning, exploration

Underlying intelligence understanding:

Ability to adapt, learn and generalize

Physical Turing Tests

- Incorporating robotics and emboddied AI
- Physical goal fulfillment
  - IKEA test
  - Coffee test
- Physical indistinguishability
  - The Physical Turing Test

Underlying intelligence understanding:

Physical presence and interaction as crucial part of intelligence

Benchmarking AGI

- Many different approaches
- Defined by underlying understanding of intelligence
- Specialized for certain AI subfields/modalities
  - Can be combined
  - But no guarantee for completeness
- No harmonized approach without shared definition

Feasibility - Timelines

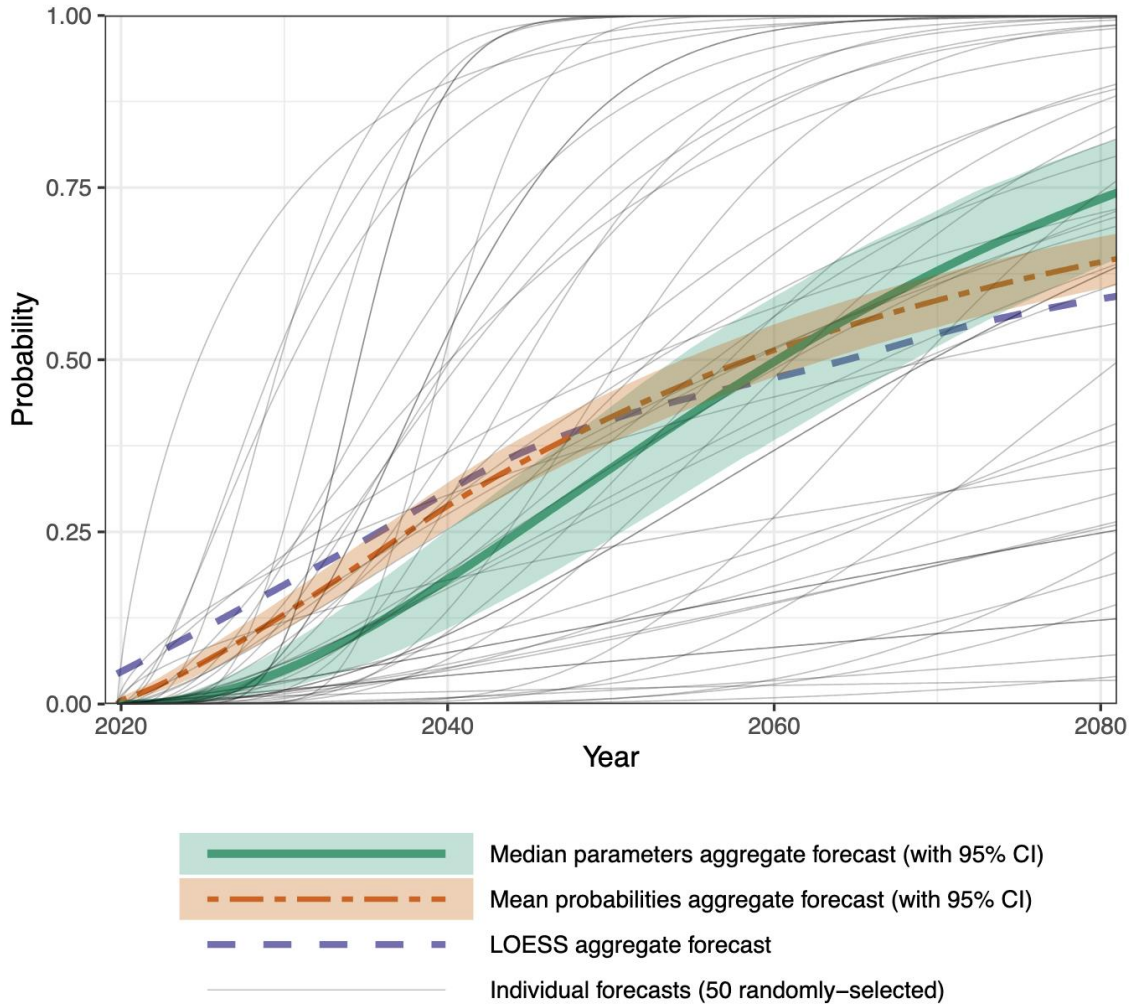
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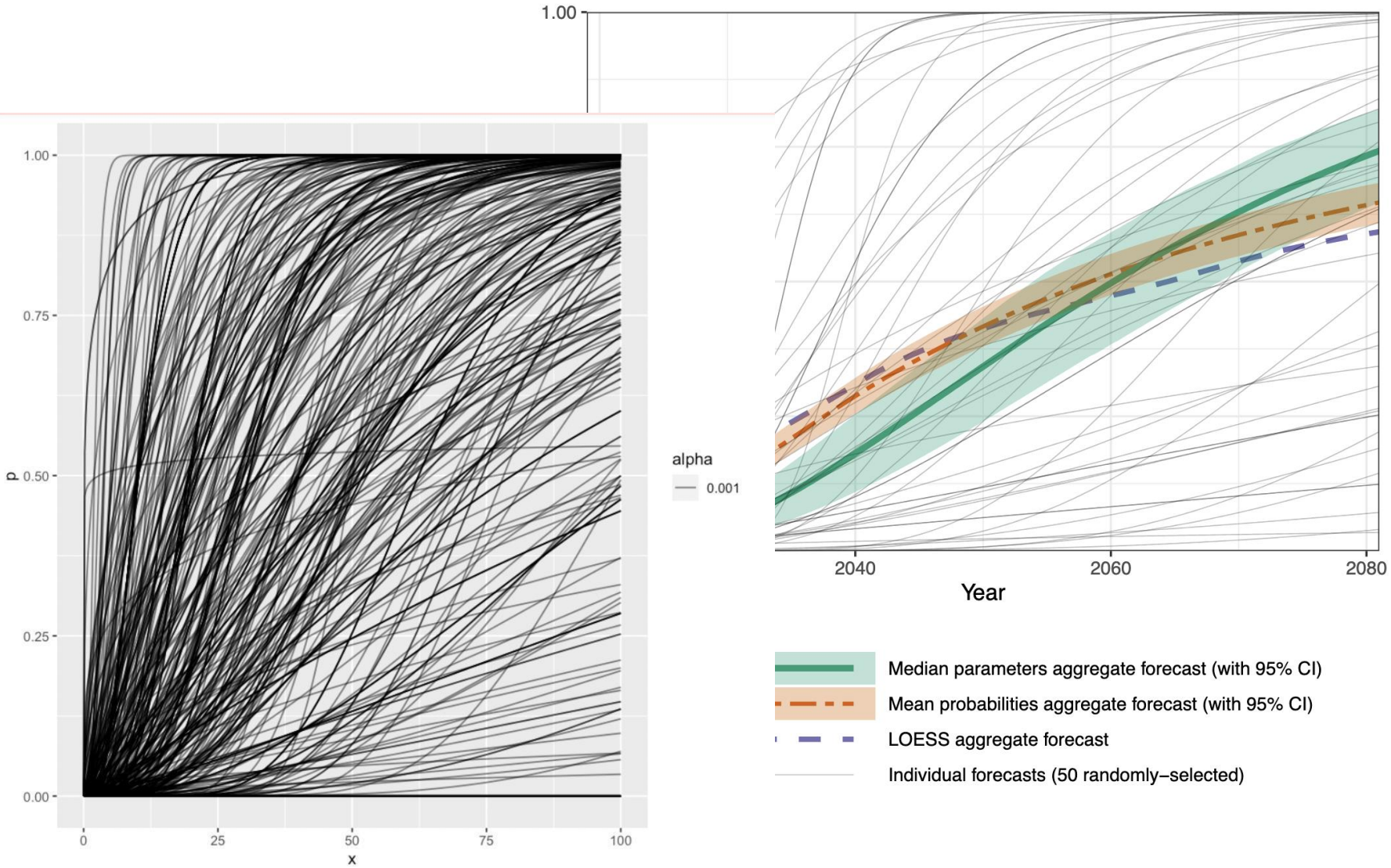
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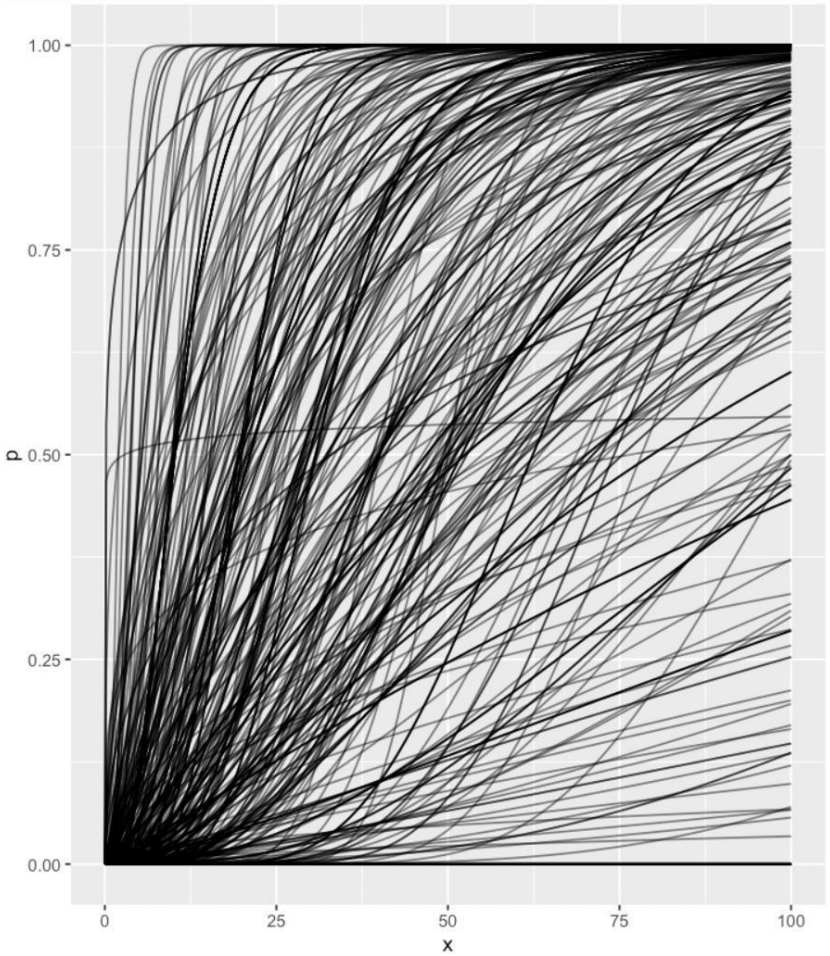
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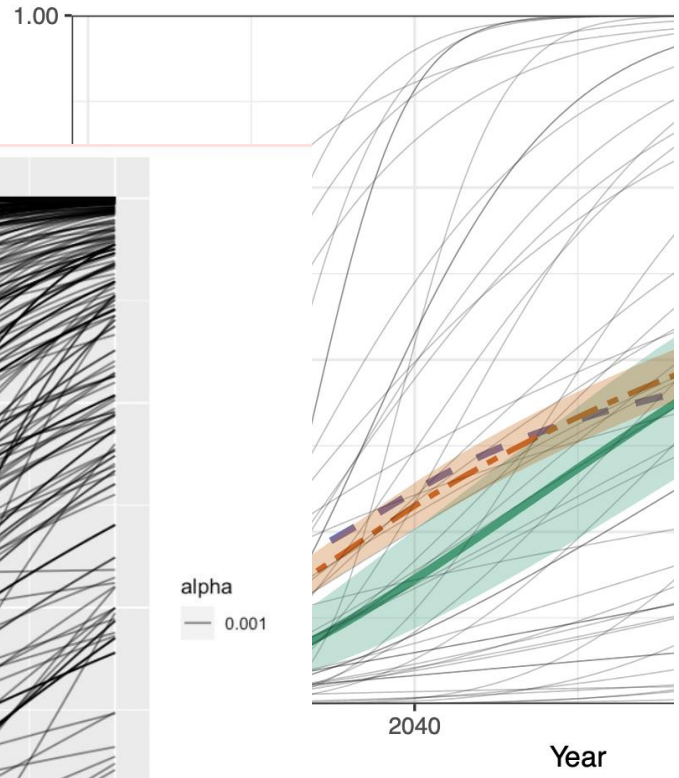


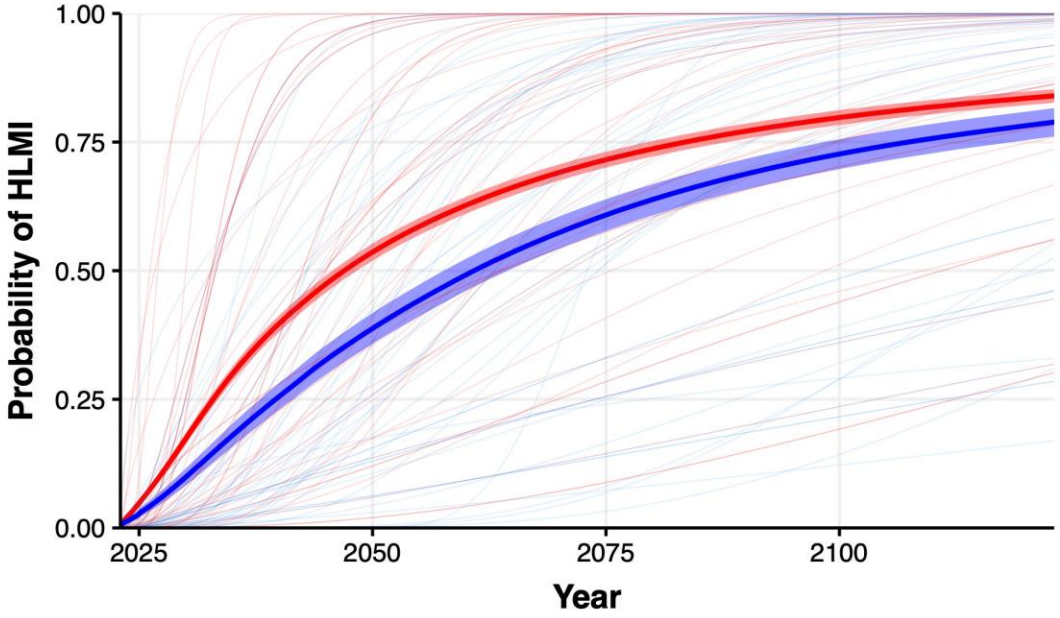
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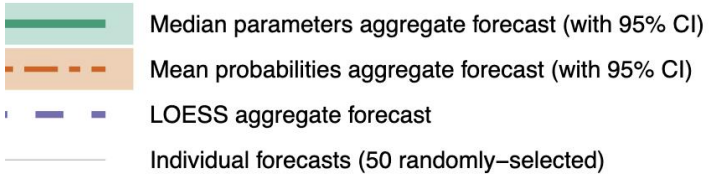


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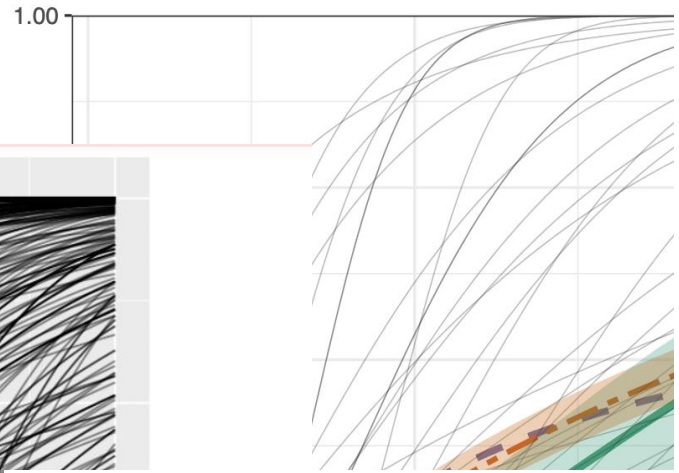


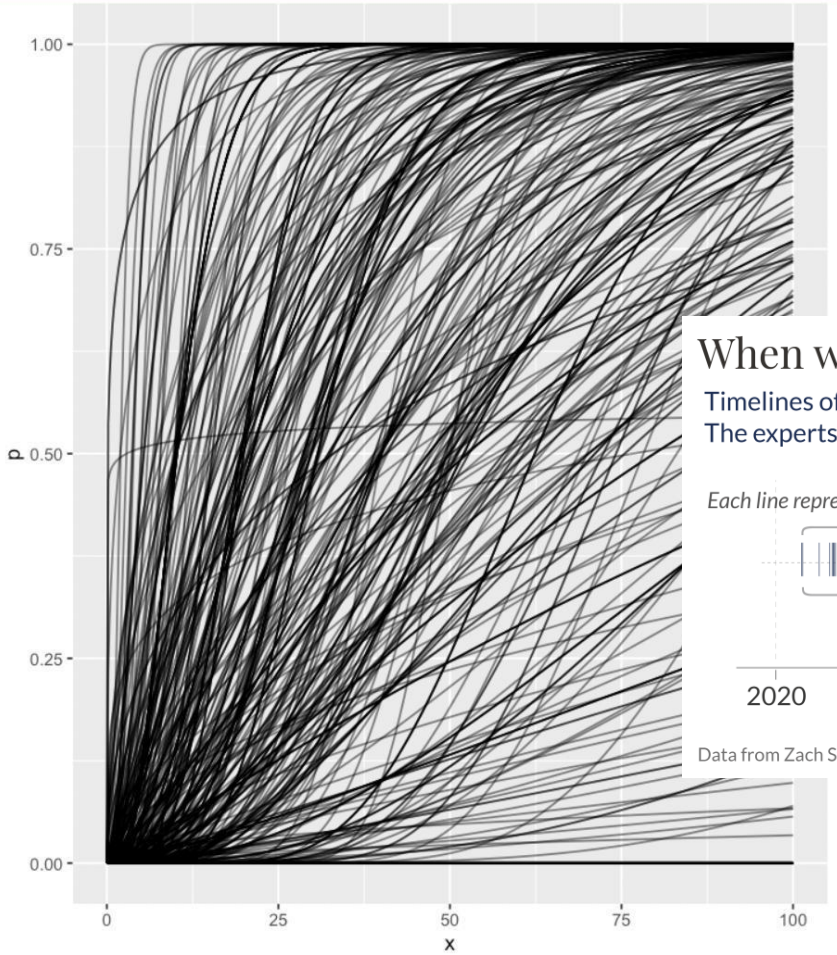


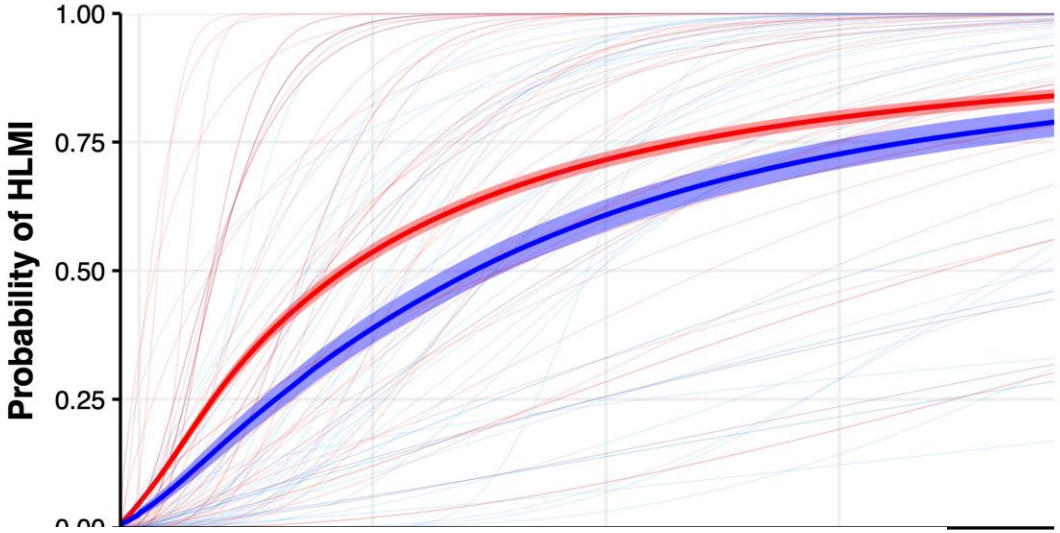


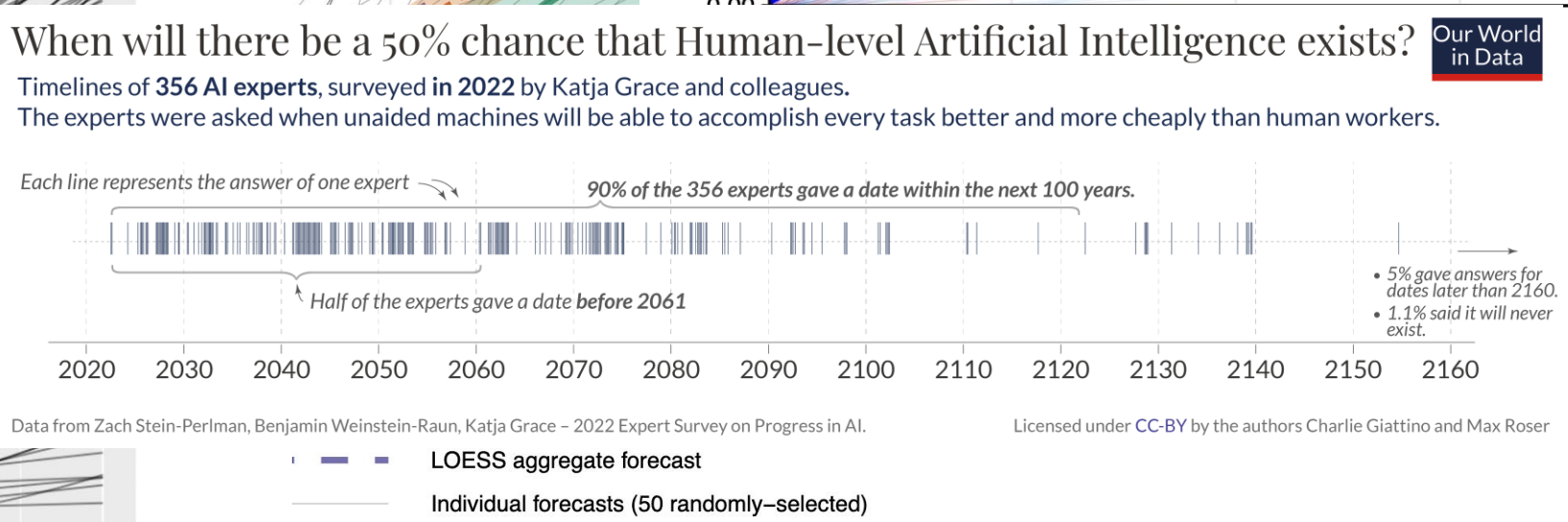


Feasibility - Timelines









Feasibility - Timelines

- AGI definitions differ
- “Experts” disagree
- Predictions change

Feasibility - Approaches

- Advancing SOTA neural networks
  - Symbolic AI
- Biologically inspired AI

Feasibility - Better NNs

The approach:

- Larger networks
  - More complex architectures
- Increasing modalities
- (New learning techniques)
- (Symbolic rpresentations)

The problem:

- Limited training data
- Very low efficiency
- No biologic foundation

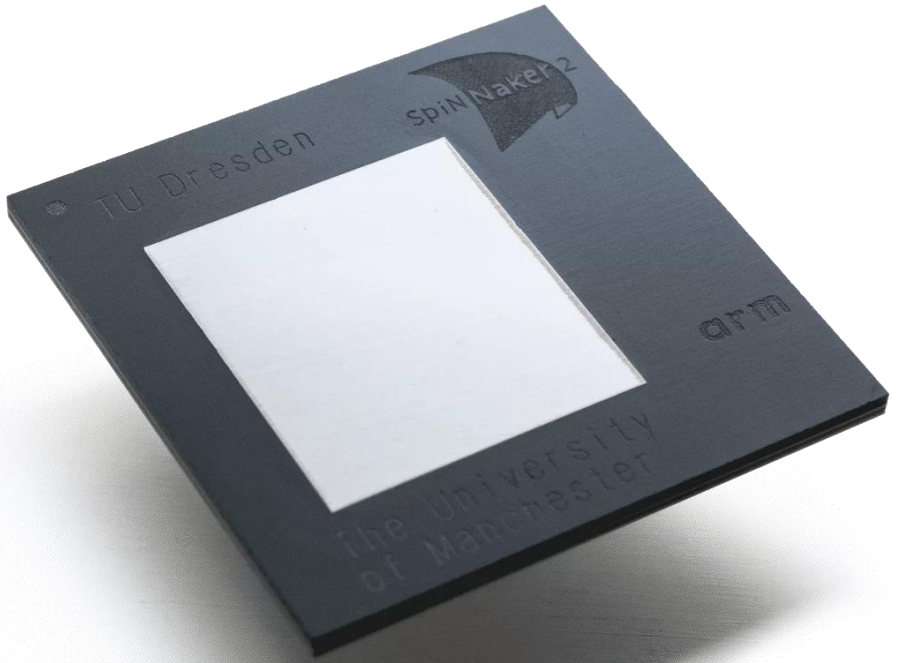
Feasibility - Bioinspiration

The approach:

- Neuromorphic computing
  - Spiking network architectures
  - Specialized hardware
- (Whole) brain emulation

The problem:

- Limited emulation power
- Very low efficiency
- Scanning resolution
- Conceptual correctness



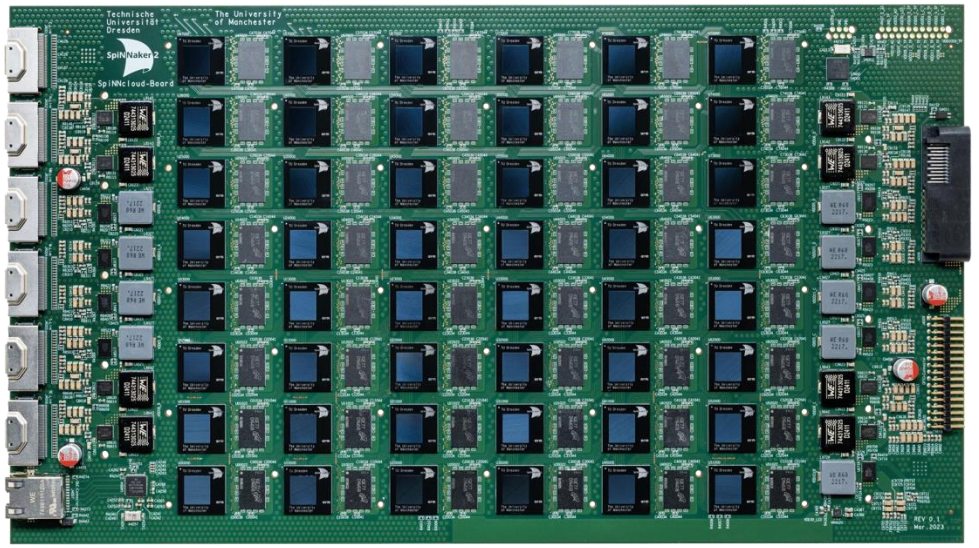
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Implications - Prediction Accuracy

- Predictions are hard to impossible
- Increasing layers of assumptions
  - What is intelligence?
  - When is AGI achieved?
  - How does it function?
  - What properties does it have?
- Spectrum of predictions
  - Few assumptions -> broad, obvious, boring, likely correct
  - Many assumptions -> specific, interesting, likely incorrect

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When encountering predictions ask yourself:

How many and which assumptions are made?

Implications - Prediction Scale

“AGI would have a big impact on several aspects of society”

Implications - Prediction Scale

"AGI would have a big impact on several aspects of society"

?

Implications - Prediction Scale

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?

- Industrial revolution or extinction level event?

Implications - Prediction Scale

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- Industrial revolution or extinction level event?
- Linear, exponential, or disruptive development?

Summary

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- Hard facts on AGI are sparse
- This kind of debate is neither new nor useless
- Important to focus on current AI developments

